Activity-9

(1) Base Map : A choropleth map shows the percentage of high-school graduates across 48 contiguous states. The highest state, lowest state & median can be identified using R.

```
max (d$ hsgrad)
d$ state [which. max (d$ hsgrad)]
min (d$ hsgrad)
d$ state [d$ hsgrad == min (d$hsgrad)]
median (v)
```

(2) Regional Story :

```
tapply (v, d$ zone, mean)
cor (v, w).
```

The regional means show differences in high-school graduation rates. correlation indicates the strength and direction of the relationship b/n population & graduation rate, but it does not prove causation

(3) Classes change the story :

```
cut (v, 4)
cut (v, quantile (v, seq (0,1, length.out=5)),
include. lowest = TRUE)
```

Equal-interval classes use equal value ranges. while quantile classes distribute states more evenly. Quantile classes can make high values appear less rare.

(4) Counts vessus sates:

```
count <- d$pop * 1000 * d$ hsgrad/100
head(d$ state [osdex (count . deoreasing = True]] . 3]
head (d$ state [osdex (v, deoreasing = True)], 3)
length (intersect (d$ state [osder (count, decreasing =
    True)] [1:3], d$ state [order (v, deoreasing = True)
        [1:3])}
```

- Raw counts may mislead because states have different populations. Mapping graduation percentages provides a faiser comparison.

(5) Nosthea st regional story:

```
n<-d$zone == "Nosthеast".
·mean (v[n])
 mean (v [n]) / mean (v).
 d$ state [n] [which . max (v[n])]
  max. (v[n]).
  sum (v[n] > median (v))
```

The Nosthеast mean can be compared with the national mean to identify regional differences. The highest nosth east state & the number of states above the national median provide supporting evidence.